COMPSCI 201 – Final Group Project

Fall 2026 (Session 1)

For this group project, students will use **Java** to create a graphics application focusing **on DKU**. The application may include but not be limited to the navigation of campus/buildings, campus life, experience/interaction with various units or departments, courses, policies, etc. – **be creative**, any application (or even a game) about DKU will do!

The **primary objective of the project** is to demonstrate your understanding and application of concepts learned in Compsci201. The complexity or visual sophistication of the application alone does not determine the grade.

The application may run in Windows or Mac computers.

Use of AI

Limited use of generative AI tools is permitted for this project as a learning and development aid. AI may be used for purposes such as enhancing the understanding of concepts, debugging assistance, or suggestions for improving code. The group remains fully responsible for all submitted work and must understand and be able to explain all submitted code.

Any use of generative AI must be declared in the project report and presentations. The declaration should identify the tool used, explain how and for what purpose it was used, and provide representative evidence of significant AI interactions. Undeclared AI use will be handled in accordance with the course/university academic-integrity policy.

Project Requirements

1. **Graphics and Interaction**: the application should have a graphical user interface (GUI). You may use Java Swing or other Java-compatible graphics/GUI framework.
2. **Data Structures**: use appropriate Java data structures (e.g., list, stack, queue, heap, tree, or map) as a meaningful part of the application. Your choice and its purpose should be explained in the report.
3. **Team Collaboration**: each group must consist of **2-3 students.** One or more than 3 students in a group is/are NOT allowed.
4. **Documentation and Presentation**: java project folder, project report and a group presentation.

Evaluation Criteria

Your group project will be graded out of 100 points based on the following:

1. Overall quality of **application** (50 points):
    a. Functionality and features (15): creativeness, richness
    b. Design and user interaction (10): clear GUI, intuitive interaction
    c. Appropriate use of data structures (15): correct use; suitable choices
    d. Code quality/readability (10): readability, appropriate classes/methods, comments.
2. Overall quality of **report** (20 points) – structure, text, figure, diagram, etc.
3. Overall quality of **presentation** (30 points) – content, flow, clarity, collaboration of team members, time management, live demo, slides design, Q/A.

Submission:

Only does **one group member** need to submit the project, not every group member.

1. **Application**:
    - Deadline of submission: **Oct 13th, 11:50PM, 2026**
    - ZIP and submit to Canvas the complete Java project folder created using VS Code, IntelliJ IDEA, or another Java development environment. Before submission, extract your ZIP file into a different folder/computer and verify that your application can be compiled and run from the submitted files. Failing to do so may result in the deduction of points.
    - Name the zip file as **X_Y.zip, or X_Y_Z.zip**, where X, Y and Z are the first names of the group members.
2. **Report**:
    - Deadline of submission: **Oct 13th, 2026**
    - Submit to Canvas a PDF file with the same name convention as that for the Application above
    - The file should be **4-5 pages long**, including:
        - An overview of the application and its goals.
        - Software design details, such as UML diagrams and class descriptions.
        - Descriptions of data structures used and their applications in the project.
        - Describe your testing approach and give at least two examples of significant bugs/problems encountered, how you surfaced them, and how they were resolved.
        - Describe the **use of AI**, as stated in the section 'Use of AI'.
        - A reflection on team collaboration. Give the percentage of each member's contributions if they differ.
3. **Presentation**:
    - Date: **Oct 15th, 2026,** during the lab time.
    - **5-minute** long, followed by a **2-minute** Q&A.
    - Do not attempt to cover the entire Report. Instead, focus on the application idea, key design/data-structure decisions, and live demonstration.
    - Include slide(s) on the **key area(s)** for which the group has not used AI.
    - Submit to Canvas the PowerPoint file used for your presentation by **Oct 13th**.